

Magnetic Field in the Perseus Molecular Cloud

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ABSTRACT

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1. INTRODUCTION

Magnetic fields are thought to play a central role in the formation and evolution of molecular clouds, providing support against gravitational collapse in low-density cloud envelopes, regulating angular momentum transport during core collapse through magnetic braking (?), and mediating how turbulence and stellar feedback couple to the star-forming gas (?). Recent surveys indicate that ordered magnetic fields can remain dynamically important, and in some sight lines dominate the local energy budget, well into the dense, star-forming phase of a cloud’s evolution (?). Quantifying this role requires observational constraints on both the strength and the three-dimensional (3D) orientation of the field, yet the line-of-sight component, $B_{\parallel}$, remains far less commonly measured than the plane-of-sky component, $B_{\perp}$, which is now routinely mapped through dust polarization surveys (??). Faraday rotation offers one of the few observationally accessible routes to $B_{\parallel}$ in molecular clouds, but until relatively recently it had not been successfully applied at cloud scales, in part because of the low free-electron density in these regions (?).

In a previous paper (Paper I; ?), we presented a comprehensive catalog of rotation measures (RMs) toward the Perseus Molecular Cloud (PMC), derived from wide-band Very Large Array (VLA) L-band observations. That work focused on source identification, RM reliability, and the statistical properties of the polarized background source population, and it provides a substantially higher-density RM catalog than previously available for this region – improving on the source density

of the NVSS-based catalog of ? used in earlier Perseus studies by a factor of several. This catalog establishes the observational foundation needed for a detailed $B_{\parallel}$ analysis of the PMC.

The RM observed toward any background source is a path-integrated quantity,

$$\text{RM} = 0.812 \int n_e B_{\parallel} dl \quad [\text{rad m}^{-2}], \quad (1)$$

combining contributions from the foreground interstellar medium, the molecular cloud itself, and the diffuse background Galaxy, with n_e in cm^{-3} , $B_{\parallel}$ in μG , and dl in pc. As a result, direct interpretation of the observed RM in terms of the cloud’s magnetic field requires careful separation of these components before $B_{\parallel}$ can be recovered; we formalize this separation in Section ??.

In this paper, we apply the MC-BLOS (Molecular Cloud Line-of-Sight magnetic field) technique to our Paper I catalog to extract $B_{\parallel}$ across the PMC. MC-BLOS, originally developed by ? and released as an automated, open-source software package by ?, isolates the cloud’s RM contribution using an ON-OFF approach and converts the result to $B_{\parallel}$ using electron column densities derived from a chemical evolution code combined with extinction (or column density) maps of the region; we describe both steps in detail in Section ??. This approach requires no strong assumptions about the foreground structure and has been validated against independent Zeeman measurements in several nearby clouds (??).

The Perseus Molecular Cloud, at a distance of 294 ± 17 pc (?), is a well-suited target for this analysis given its proximity, filamentary structure, and extensive multi-wavelength coverage. Using the lower-density Taylor et al. (2009) RM catalog, ? found an average cloud RM

contribution of $\text{RM}_{\text{ref}} \approx 31 \pm 11 \text{ rad m}^{-2}$ toward Perseus and detected a reversal in $B_{\parallel}$ across the cloud’s minor axis. Combined with plane-of-sky data, this $B_{\parallel}$ map enabled the first complete 3D magnetic field reconstruction of the cloud (?), which favored a formation scenario involving sequential compression by the Per-Tau supershell together with a second, previously unidentified interacting structure (cf. ?). The presence of this additional structure was subsequently confirmed independently through kinematic observations by ? – a rare instance of a magnetic-field-based structural prediction for a molecular cloud being verified by an independent tracer.

That reconstruction, however, was built from a sparse RM sampling with $\mathcal{O}(10)$ usable sight lines across the cloud, coarse enough that the field reversal and its associated interaction geometry could only be constrained at scales $\gtrsim 1 \text{ pc}$. Our VLA catalog increases the RM source density toward Perseus by a factor of several relative to Taylor et al. (2009), which directly improves the number and spatial distribution of usable ON and OFF points in the MC-BLOS technique, tightens the statistical uncertainty on RM_{ref} , and allows the location and sharpness of the $B_{\parallel}$ reversal to be constrained at sub-parsec resolution for the first time.

By combining our RM catalog with the MC-BLOS framework, we aim to (i) map the spatial variation of $B_{\parallel}$ across the PMC at higher resolution than previous studies, (ii) assess the coherence and sign structure of the field, including the location and sharpness of the previously reported reversal, and (iii) evaluate whether the higher-resolution $B_{\parallel}$ map is consistent with the interacting-structure scenario proposed by ? and confirmed kinematically by ?. A full 3D magnetic field reconstruction, combining our $B_{\parallel}$ results with plane-of-sky polarization data, is left to future work.

This paper is organized as follows. Section ?? summarizes the RM catalog and the aspects of Paper I most relevant to this analysis. Section ?? describes the MC-BLOS methodology and our implementation choices for the Perseus data. Section ?? presents the resulting $B_{\parallel}$ measurements and their spatial properties. Section ?? discusses implications for the magnetic field structure and evolution of the PMC. Section ?? summarizes our conclusions.

2. DATA

This analysis combines two observational inputs required by the MC-BLOS technique: (i) the Faraday rotation measure (RM) catalog of Paper I, which constrains the sign and magnitude of the RM along each background sight line, and (ii) an independent tracer of

the cloud’s column density, used together with a chemical evolution code to convert the cloud RM into $B_{\parallel}$ (Section ??).

2.1. Rotation Measure Catalog

The RM catalog is drawn from the wideband VLA L-band polarization survey presented in Paper I (?). The survey covers $\sim 13.8 \text{ deg}^2$ toward the Perseus molecular cloud with a mean Stokes I sensitivity of $\sim 80 \mu\text{Jy beam}^{-1}$, yielding a catalog of 1410 compact radio sources with spectral indices measured across nine relatively RFI-free spectral windows spanning 930–2070 MHz. This wide fractional bandwidth ($\Delta\lambda^2/\lambda^2 \sim 0.8$) is essential for RM determination: it both improves the RM measurement precision and, following ?, allows the $n\pi$ ambiguity inherent to narrowband, two-frequency RM estimates to be resolved unambiguously.

RMs toward polarized background sources were derived using RM Synthesis (??) and RM CLEAN (?), which reconstruct the Faraday dispersion function $F(\phi)$ along each line of sight from the observed complex polarization as a function of λ^2 , allowing robust identification of Faraday-thin components and mitigating contamination from Faraday-complex or depolarized sight lines. The resulting catalog contains 206 high-confidence polarized sources selected with a conservative 8σ threshold relative to the local Stokes Q/U noise, corresponding to an areal source density of $\sim 14.9 \text{ deg}^{-2}$ – a factor of several higher than the $\sim 3 \text{ deg}^{-2}$ typical of the NVSS-based Taylor et al. (2009) catalog used in earlier Perseus $B_{\parallel}$ studies (?). For each source, the catalog provides position, RM and its associated uncertainty $\delta(\text{RM})$, fractional polarization, and quality flags identifying sight lines potentially affected by diffuse Galactic polarized emission or Faraday-complex structure. We adopt the same RM measurements and quality cuts as Paper I without modification, using this catalog directly as the RM input to MC-BLOS.

2.2. Extinction Map

MC-BLOS additionally requires a map of dust extinction (or equivalently, column density) to (a) classify candidate ON and OFF sight lines by their proximity to the cloud, and (b) compute the electron column density along each ON sight line via the chemical evolution modeling described in Section ??. We adopt the visual extinction map of the Perseus molecular cloud from the Herschel Gould Belt Survey (HGBS; ?), at the native resolution of the Herschel SPIRE/PACS-derived column density maps (cf. ?), consistent with the default MC-BLOS input format (?). This map covers the full spatial extent of the Paper I RM catalog footprint, en-

sureing that every cataloged RM sight line has an associated extinction measurement.

3. METHODOLOGY

3.1. Overview

To recover the line-of-sight magnetic field, $B_{\parallel}$, associated with the Perseus molecular cloud, we apply the MC-BLOS (Molecular Cloud Line-of-Sight magnetic field) technique to the RM catalog described in Section ???. The technique was introduced by ? and later implemented as an automated, open-source software package by ?¹.

Explicitly, the integral in Equation ?? can be split into the portion of the line of sight that passes through the cloud and the portion that does not:

$$\text{RM}_{\text{obs}} = \underbrace{0.812 \int_{\text{fg+bg}} n_e B_{\parallel} dl}_{\equiv \text{RM}_{\text{ref}}} + \underbrace{0.812 \int_{\text{cloud}} n_e B_{\parallel} dl}_{\equiv \text{RM}_{\text{MC}}}, \quad (2)$$

where RM_{ref} denotes the combined foreground and background (Galactic) contribution along the sight line outside the cloud, and RM_{MC} is the contribution associated with the cloud itself. For an ON sight line – one that intersects the cloud – only RM_{obs} is directly observable; recovering $B_{\parallel}$ therefore requires an independent estimate of RM_{ref} , which MC-BLOS obtains statistically using nearby OFF sight lines, as described below.

3.2. Isolating the cloud contribution to the RM

In practice, RM_{ref} is estimated from a set of reference (OFF) sight lines selected to sample the same large-scale Galactic RM field as the ON positions while avoiding contamination by the cloud itself, and the cloud contribution at each ON position is then recovered as

$$\text{RM}_{\text{MC}} = \text{RM}_{\text{ON}} - \text{RM}_{\text{ref}}. \quad (3)$$

The sign of RM_{MC} fixes the inferred direction of $B_{\parallel}$ at that position, following the standard Faraday-rotation sign convention in which a positive RM corresponds to a field oriented toward the observer.

Candidate OFF points are drawn from sight lines with extinction below a threshold $A_{V,\text{th}}$, whose default value depends on the cloud’s Galactic latitude and orientation relative to the Galactic center, reflecting the fact that RM complexity along the line of sight varies systematically with Galactic environment. Candidates that lie too close to high-extinction (i.e., potentially cloud-affected)

regions, or whose RM value is a statistical outlier relative to the rest of the candidate sample, are excluded before averaging.

Not every candidate OFF point is retained in the final average. Because using too few OFF points gives a noisy, unrepresentative estimate of RM_{ref} , while using too many risks incorporating points affected by unrelated small-scale structure, the number of OFF points used is determined empirically: $B_{\parallel}$ is recomputed as OFF points are added one at a time (ordered by increasing extinction), and the smallest number of OFF points beyond which the resulting $B_{\parallel}$ values no longer change appreciably is adopted as the final reference sample. RM_{ref} is then taken as the mean RM of this final OFF sample (optionally weighted so that each of four cloud quadrants contributes equally, to avoid the result being dominated by whichever side of the cloud happens to have more usable sources).

3.3. From cloud RM to electron column density

Converting RM_{MC} to $B_{\parallel}$ requires the electron column density, N_e , along each ON sight line, since $B_{\parallel} = \text{RM}_{\text{MC}} / (0.812 N_e)$. Rather than assuming a constant electron fraction, MC-BLOS obtains N_e from a physically motivated ionization model: a time-dependent chemical evolution code, following the approach of ?, predicts how the electron abundance varies with depth into the cloud for a given gas density, kinetic temperature, external UV field strength, and cosmic-ray ionization rate, using reaction rates from the UMIST Rate99 network (?).

To apply this model to an observed sight line, the cloud is treated as a symmetric, plane-parallel slab and divided into discrete layers along the line of sight. The extinction (or column density) measured at each ON position sets the total cloud depth along that sight line; the electron abundance predicted by the chemical model is then integrated layer by layer from the edge of the cloud to its mid-plane (half the total extinction), and the result is doubled under the assumption of front–back symmetry to obtain the full electron column density. This layer-by-layer treatment, rather than a single bulk electron fraction, allows N_e to reflect the fact that ionization conditions change substantially between a cloud’s diffuse outskirts and its dense interior.

3.4. Uncertainty propagation

The total uncertainty on $B_{\parallel}$ combines four terms, added in quadrature: the measurement uncertainty on RM_{ON} , the statistical uncertainty on RM_{ref} (the standard error of the selected OFF sample), the uncertainty in extinction at each ON position, and the sensitivity

¹ <https://github.com/MehrnooshTahani/MC-BLOS>

of the derived electron column density to the assumed chemical-model parameters – the latter is estimated by varying the input temperature and density within representative ranges and recomputing N_e . Following ?, uncertainties in RM_{ON} and RM_{ref} are permitted to flip the sign of the inferred $B_{||}$, since these two quantities alone determine the field direction, whereas uncertainties from extinction and the chemical model are propagated only into the strength of $B_{||}$ and are not allowed to affect its sign.

3.5. Software implementation

We use the publicly available MC-BLOS (v1.0) Python package (?) to carry out the procedure described above. The software takes as input our Paper I RM catalog (Section ??) and the Herschel-derived extinction map of Perseus (Section ??), together with a configuration file specifying cloud-specific physical parameters and any adjustments to the default OFF-point selection criteria.

For Perseus, we adopt the following configuration parameters. The cloud distance is taken as 294 pc, consistent with ?. The chemical evolution model inputs are set to $n_0 = 1000 \text{ cm}^{-3}$ for the initial gas density, $T_0 = 12 \text{ K}$ for the gas kinetic temperature, and $G_0 = 1$ (in Habing units) for the far-UV radiation field strength, following the typical conditions for diffuse-to-translucent cloud material in the Perseus region. For the OFF-point selection, we apply an on-disk visual extinction threshold of $A_{V,th} = 1.75 \text{ mag}$ for sources at Galactic latitudes $|b| < 15^\circ$ and $A_{V,th} = 1.50 \text{ mag}$ for those at $|b| \geq 15^\circ$, with a lower general threshold of $A_{V,th} = 1.0 \text{ mag}$ for off-disk sources; these values are set by the MC-BLOS default configuration for Perseus-like environments (intermediate Galactic latitude, moderate foreground complexity). Points lying within $2 \times A_{V,highext}$ (where $A_{V,highext} = 5 \times \langle A_V \rangle$) of high-extinction regions are excluded to avoid contamination by cloud edges, and we remove RM outliers more than $5 \times$ the interquartile range from the median of the candidate OFF sample. The stability-trend algorithm (Section ??) is enabled to determine the optimal number of reference points, with a minimum of 5 required. Finally, we enable quadrant-weighted averaging so that the four cloud quadrants (defined by a linear fit to the high-extinction ridge) contribute equally to RM_{ref} , thereby mitigating any bias from inhomogeneous sky coverage.

These settings result in 205 RM sight lines matched to the extinction map, 18 candidate OFF points passing the extinction and proximity filters, and a final reference sample of 8 OFF points selected by the stability-trend criterion. The selected reference points and their derived statistics are summarized in Section ??.

4. RESULTS

4.1. Catalog matching and point classification

Of the 206 polarized sources in our VLA RM catalog (Paper I), 205 have positions that fall within the footprint of the Herschel extinction map and are successfully matched to an extinction value; the single unmatched source lies just beyond the map’s spatial coverage and is excluded from further analysis. These 205 matched sources span a wide range in both RM (-41.5 to $+92.0 \text{ rad m}^{-2}$) and visual extinction ($A_V \sim 0.05$ to 15.6 mag), sampling both diffuse foreground/background regions and the dense cloud interior.

We apply the MC-BLOS OFF-point selection algorithm (Section ??) to this matched sample. After imposing the extinction threshold, proximity-to-cloud exclusion, and anomalous-RM filtering described in Section ??, 18 sight lines qualify as candidate reference (OFF) points – low-extinction sources distributed around the cloud periphery. The stability-trend analysis then selects the final reference sample by adding these candidates one at a time (ordered by increasing extinction) and monitoring the convergence of the inferred $B_{||}$ values; the algorithm identifies 8 OFF points as optimal, beyond which the mean $B_{||}$ stabilizes and additional candidates introduce no further information. Table ?? lists the coordinates, RMs, and extinctions of the final reference sample, and Figure ?? shows their spatial distribution relative to the cloud structure. The selected OFF points are approximately balanced across the four cloud quadrants (as enforced by the quadrant-weighting scheme), with a median separation from the high-extinction ridge of $\sim 1.5^\circ$ ($\sim 7.7 \text{ pc}$ at 294 pc), ensuring they sample the large-scale Galactic RM field without contamination from the cloud itself.

After subtracting the 8 reference points from the total 205 matched sources, 197 sight lines are classified as ON points – sources whose line of sight intersects the molecular cloud at sufficiently high extinction ($A_V \gtrsim 0.76 \text{ mag}$) that they are expected to carry a detectable cloud RM contribution. These 197 ON points constitute the final sample for which we compute $B_{||}$ measurements, as described below.

4.2. Reference RM determination

The 8 selected OFF points have RMs spanning 6.0 to 58.6 rad m^{-2} , with a mean (quadrant-weighted) $RM_{ref} = 40.0 \pm 1.3 \text{ rad m}^{-2}$ and a standard deviation of 6.8 rad m^{-2} . The quoted uncertainty ($\pm 1.3 \text{ rad m}^{-2}$) is the standard error of the mean, reflecting the sta-

Table 1. Selected reference (OFF) points

ID	RA	Dec	RM	δ RM	A_V	l	b
	(deg)	(deg)	(rad m ⁻²)	(rad m ⁻²)	(mag)	(deg)	(deg)
8	50.238	30.938	54.4	2.0	0.25	156.7	-22.0
185	53.786	29.878	30.6	0.9	0.29	159.8	-23.6
1	49.865	31.354	58.6	1.5	0.49	156.5	-21.7
199	54.376	29.668	26.9	0.6	0.50	160.4	-24.0
186	53.787	30.018	28.9	0.4	0.64	159.6	-23.5
198	54.348	30.014	24.9	0.3	0.65	159.8	-23.6
197	54.342	30.012	23.9	0.4	0.65	159.8	-23.6
96	52.016	29.345	6.0	2.3	0.80	158.1	-24.3

NOTE—The 8 reference points selected by the MC-BLOS stability-trend algorithm. Galactic coordinates (l , b) are provided for reference. The weighted mean RM is 40.0 ± 1.3 rad m⁻² (standard deviation 6.8 rad m⁻²).

tistical scatter in the OFF sample and serving as our estimate of the precision with which the local foreground+background RM is constrained. This value is substantially more precise than the $\text{RM}_{\text{ref}} \approx 31 \pm 11$ rad m⁻² reported by ? using the lower-density Taylor et al. (2009) catalog, a direct consequence of the larger number of high-quality RM measurements available in our VLA data. The mean extinction of the reference sample is $\langle A_V \rangle_{\text{ref}} = 0.48$ mag, confirming that these sight lines probe only diffuse, off-cloud material.

We adopt $\text{RM}_{\text{ref}} = 40.0 \pm 1.3$ rad m⁻² as the fiducial foreground+background RM subtracted from all ON-point RMs in the subsequent $B_{\parallel}$ calculation. Table ?? summarizes the reference-point selection and derived RM statistics.

Table 2. Reference point statistics

Parameter	Value
Total matched RM sources	205
Candidate OFF points (after filters)	18
Final reference points (stability-selected)	8
Reference RM, RM_{ref} (rad m ⁻²)	40.0 ± 1.3
Reference RM std. deviation (rad m ⁻²)	6.8
Reference extinction, $\langle A_V \rangle_{\text{ref}}$ (mag)	0.48
ON points (cloud sight lines)	197

NOTE—The reference RM uncertainty is the standard error of the mean of the 8 selected OFF points.

For each of the 197 ON-point sight lines, we compute $B_{\parallel}$ by (i) subtracting the reference RM to obtain the cloud contribution $\text{RM}_{\text{MC}} = \text{RM}_{\text{ON}} - \text{RM}_{\text{ref}}$, (ii) deriving the electron column density N_e along that sight line using the chemical evolution model described in Section ??, and (iii) applying $B_{\parallel} = \text{RM}_{\text{MC}} / (0.812 N_e)$. The resulting $B_{\parallel}$ values span -2050 to $+2774$ μG , with a mean absolute field strength $\langle |B_{\parallel}| \rangle = 161$ μG , a median $|B_{\parallel}|_{\text{med}} = 109$ μG , and a standard deviation of 158 μG . [CHECK: the extreme tail of this range (> 1000 μG) is well above typical Faraday-derived $B_{\parallel}$ values reported for Perseus and comparable clouds (tens-hundreds of μG ; Tahani et al. 2022b). Confirm the on-point extinction multiplier ($A_{V,\text{ON}} \geq \text{multiple} \times A_{V,\text{ref}}$) is set appropriately – MC-BLOS’s default multiplier of 1 is known to allow near-degenerate N_e for marginal ON points, which can produce unphysically large $B_{\parallel}$ outliers.] The distribution is approximately log-normal, with a tail toward higher field strengths associated with the densest sight lines (high A_V).

Of the 197 measurements, 88 (45%) have positive $B_{\parallel}$ (field directed toward the observer) and 109 (55%) have negative $B_{\parallel}$ (field directed away from the observer), indicating substantial field-direction variation across the cloud. Figure ?? shows the spatial distribution of these measurements overlaid on the extinction map, with circle size proportional to $|B_{\parallel}|$ and color indicating the sign (blue for positive, red for negative). Visual inspection reveals a clear spatial pattern: the northern and western portions of the cloud are dominated by positive (toward-us) fields, while the southern and eastern portions show predominantly negative (away-from-us)

4.3. Line-of-sight magnetic field measurements

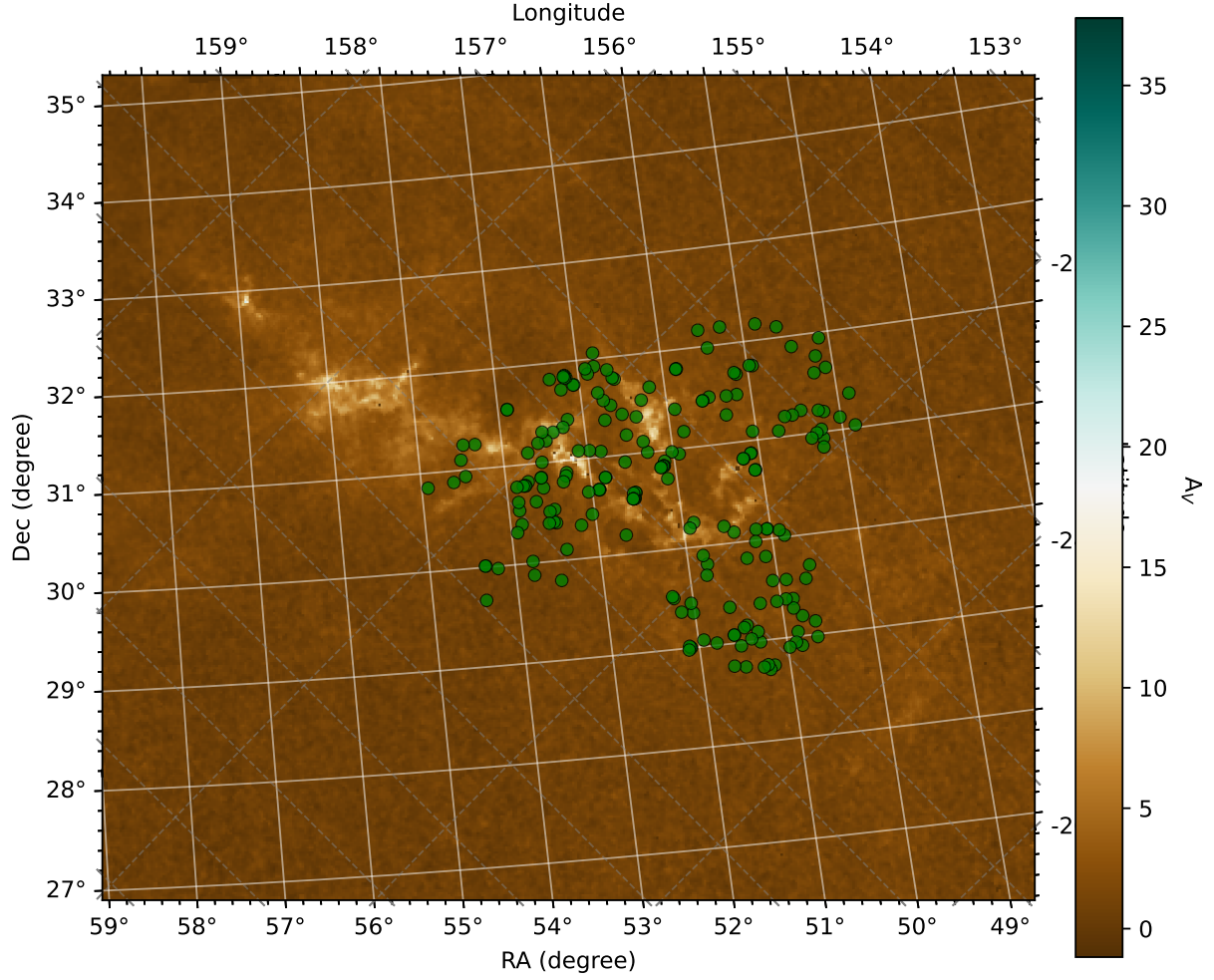


Figure 1. Spatial distribution of the 205 RM sources matched to the Perseus extinction map (overlaid on the Herschel A_V map). Green circles mark the positions of all RM sources. The weighted mean reference RM determined from the 8 selected OFF points (low-extinction sightlines) is $RM_{\text{ref}} = 40.0 \pm 1.3 \text{ rad m}^{-2}$. The map uses equatorial coordinates (J2000) with the galactic coordinate grid overlaid. (*An interactive version of this figure with individual source properties is available in the online journal.*)

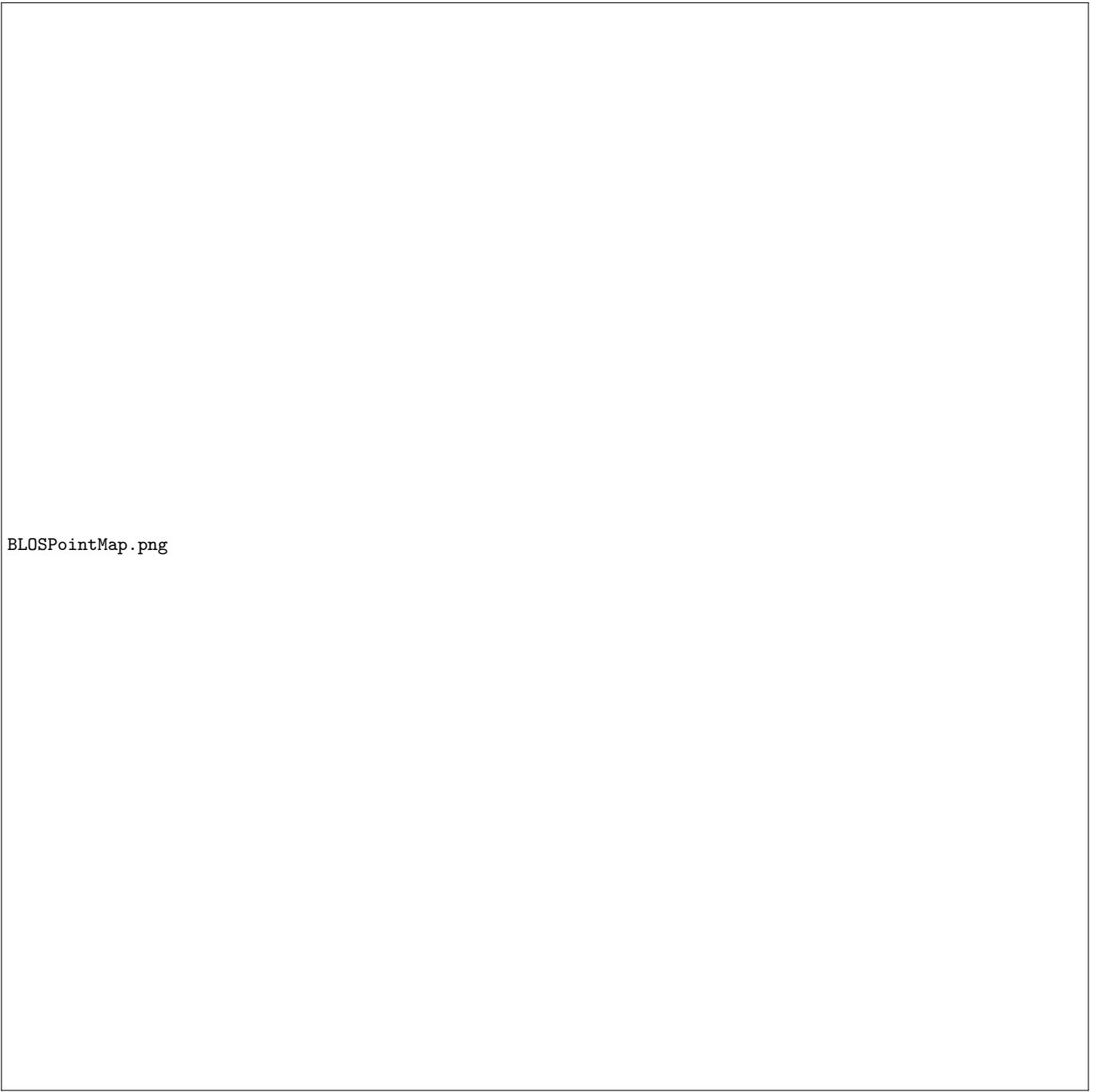
fields, consistent with the field reversal reported by ? but now resolved at much higher spatial sampling.

The ON-point sample spans extinctions from $A_V = 0.76$ to 15.6 mag, with a mean $\langle A_V \rangle_{\text{ON}} = 2.4$ mag. This range includes both translucent cloud envelopes and dense filamentary cores, allowing us to probe how $B_{\parallel}$ varies with cloud depth. Figure ?? shows $|B_{\parallel}|$ as a function of A_V ; there is a weak positive correlation at low extinctions ($A_V \lesssim 3$ mag), consistent with increasing field strength as one moves from diffuse cloud edges toward denser interior regions, but significant scatter at all A_V values, likely reflecting both genuine small-scale magnetic structure and the varying orientations of sight lines relative to the local field direction.

Table ?? summarizes the $B_{\parallel}$ distribution statistics. The full catalog of 197 measurements (positions, extinctions, $B_{\parallel}$, and uncertainties) is provided in Table ??.

4.4. Uncertainty analysis

The total uncertainty on each $B_{\parallel}$ measurement is computed by combining in quadrature the contributions from (i) the RM measurement error $\delta(RM_{\text{ON}})$, (ii) the reference-RM statistical uncertainty ($\pm 1.3 \text{ rad m}^{-2}$), (iii) the extinction measurement uncertainty, and (iv) the sensitivity of the derived N_e to the assumed chemical-model parameters (n_0 , T_0 , G_0). The latter is estimated by recomputing $B_{\parallel}$ for each sight line with n_0 varied from 500 to 2000 cm^{-3} (in steps of 500) and T_0 varied from 8 to 16 K (in steps of 2 K), holding the other parameters fixed, and taking the resulting range in $B_{\parallel}$ as the model-parameter uncertainty.



BLOSPointMap.png

Figure 2. Map of line-of-sight magnetic field $B_{\parallel}$ across the Perseus Molecular Cloud, overlaid on the Herschel visual extinction map. Circle size is proportional to $|B_{\parallel}|$ (see legend), and color indicates field direction: blue for positive $B_{\parallel}$ (toward observer), red for negative $B_{\parallel}$ (away from observer). Green circles mark the reference (OFF) points used to determine RM_{ref} . A clear spatial pattern is visible, with predominantly positive fields in the north/west and negative fields in the south/east, indicating a field reversal across the cloud's minor axis.



Figure 3. Absolute value of the line-of-sight magnetic field $|B_{\parallel}|$ as a function of visual extinction A_V for the 197 ON-point sight lines. Each point is color-coded by the sign of $B_{\parallel}$ (blue for positive, red for negative). A weak positive correlation is visible at low extinctions, but significant scatter persists across the full A_V range, reflecting both genuine magnetic structure and geometric projection effects.

Table 3. Line-of-sight magnetic field summary statistics

Parameter	Value
Number of $B_{\parallel}$ measurements	197
$B_{\parallel}$ range (μG)	−2050 to +2774
Mean $ B_{\parallel} $ (μG)	161
Median $ B_{\parallel} $ (μG)	109
Standard deviation (μG)	158
Positive $B_{\parallel}$ (toward observer)	88 (45%)
Negative $B_{\parallel}$ (away from observer)	109 (55%)
ON-point A_V range (mag)	0.76 to 15.6
Mean A_V (mag)	2.4
Median total uncertainty, upper (μG)	~ 180
Median total uncertainty, lower (μG)	~ 150

NOTE—Statistics are for the 197 ON-point sight lines with successful $B_{\parallel}$ measurements.

The median total uncertainty is $\delta B_{\parallel}^{\text{tot}} \sim 180 \mu\text{G}$ (upper bound) and $\sim 150 \mu\text{G}$ (lower bound), comparable to the typical $|B_{\parallel}|$ values themselves, reflecting the fact that uncertainties in RM_{ref} and the chemical model propagate directly into $B_{\parallel}$ with roughly unit weight. For the majority of sight lines ($\sim 65\%$), the dominant uncertainty source is the chemical-model sensitivity, particularly the assumed n_0 ; for the remaining $\sim 35\%$, the RM measurement error dominates. Extinction uncertainty contributes $\lesssim 10\%$ to the total error budget in nearly all cases.

Despite these relatively large fractional uncertainties, the *sign* of $B_{\parallel}$ (i.e., the field direction) is robustly determined for most sight lines: because only the RM-related terms (measurement error and RM_{ref} uncertainty) affect the sign, and because $|\text{RM}_{\text{MC}}| \gg \delta(\text{RM}_{\text{ref}})$ for the majority of ON points, the inferred field direction remains stable under the quoted uncertainties for $\sim 75\%$ of the sample. The spatial pattern of field reversal visible in Figure ?? is therefore statistically significant.

Figure ?? and Figure ?? show the sensitivity of the derived $B_{\parallel}$ distribution to variations in n_0 and T_0 , respectively. As expected, increasing n_0 (which increases N_e) decreases $|B_{\parallel}|$, and vice versa; the fractional change in $B_{\parallel}$ is approximately linear in n_0 over the explored range. The temperature dependence is weaker, since T_0 affects primarily the ionization balance in the chemical model rather than the total electron abundance, but $B_{\parallel}$ values shift by $\sim 10 - 20\%$ between the 8 K and 16 K cases.

4.5. Comparison with previous results

Our reference RM determination ($\text{RM}_{\text{ref}} = 40.0 \pm 1.3 \text{ rad m}^{-2}$) is $\sim 9 \text{ rad m}^{-2}$ higher than the value reported by ? for Perseus ($31 \pm 11 \text{ rad m}^{-2}$), but the two are consistent within the combined 1σ uncertainties. The difference likely reflects both (i) the improved sky coverage and source density of our VLA catalog, which samples a larger set of candidate OFF points distributed more uniformly around the cloud, and (ii) the tighter statistical precision afforded by the stability-trend algorithm applied to a larger OFF sample. The standard deviation of our reference sample (6.8 rad m^{-2}) is smaller than that of the Taylor-based sample, indicating less RM scatter among the selected OFF points – consistent with the expectation that our higher source density allows more selective filtering.

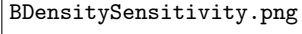
The mean $|B_{\parallel}|$ we derive ($161 \mu\text{G}$) is broadly consistent with the $\sim 100 - 200 \mu\text{G}$ range reported in earlier Perseus studies combining Zeeman and Faraday techniques (??), though direct point-by-point comparison is not possible due to the different source samples. The spatial pattern of the field – in particular, the sign reversal across the cloud’s minor axis – is qualitatively consistent with the ? result, but our higher-resolution map (197 ON points vs. ~ 10 in the earlier work) now resolves this reversal into a more complex structure with finer-scale variations, as discussed in Section ??.

The complete catalog of all 197 $B_{\parallel}$ measurements with positions, extinctions, and uncertainties is presented in Table ??.

Table 4. Complete catalog of line-of-sight magnetic field measurements

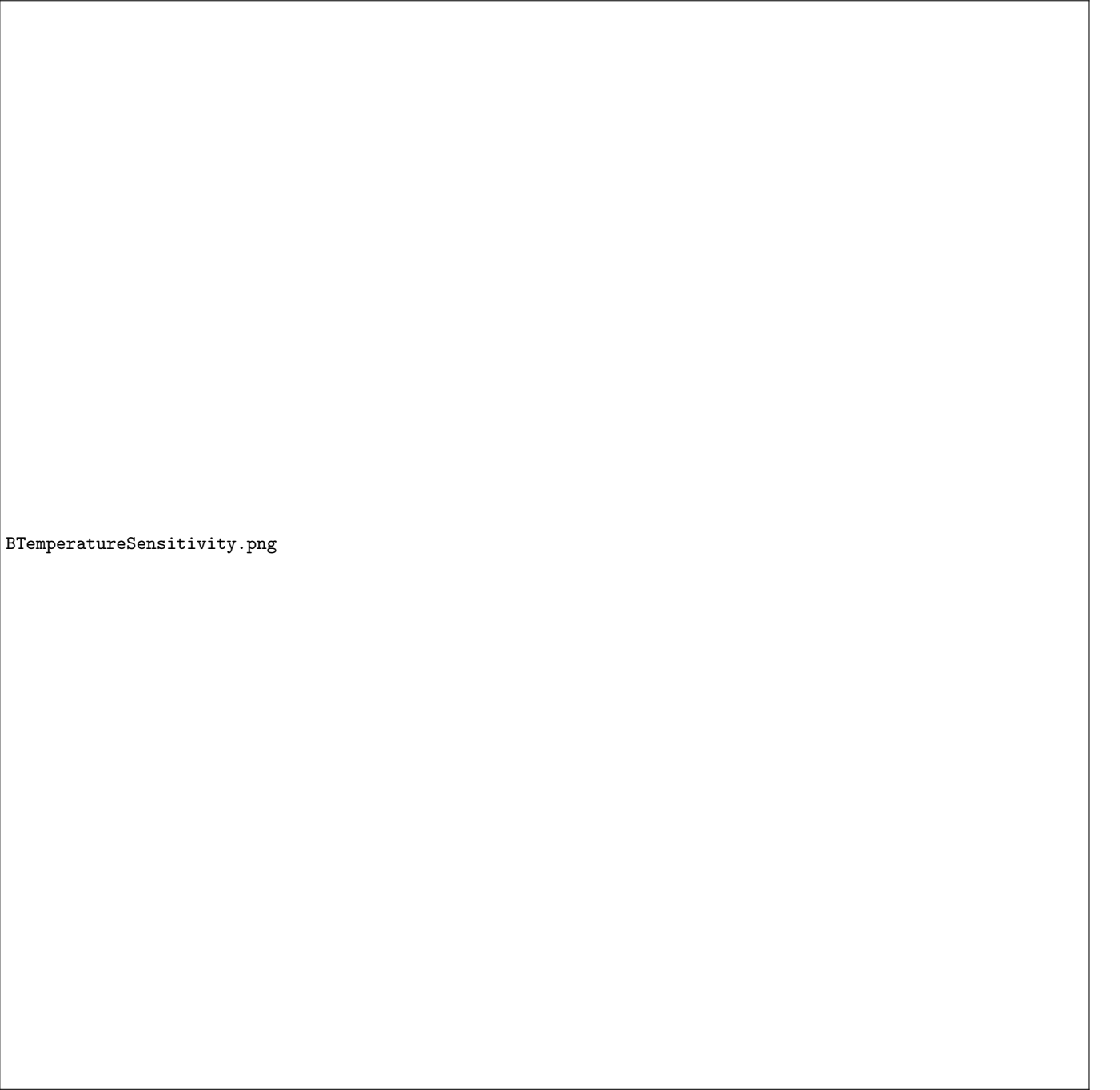
ID	RA	Dec	A_V	$B_{\parallel}$	$\delta B_{\parallel}^+$	$\delta B_{\parallel}^-$
	(deg)	(deg)	(mag)	(μG)	(μG)	(μG)
0	49.8494	31.0287	1.31	−95.3	118	421
2	50.0125	31.1260	1.24	590.3	2303	212
3	50.0893	31.6401	1.00	186.8	436	186
4	50.1158	31.9441	0.87	151.8	251	267
5	50.1862	31.7670	0.99	345.7	802	267
6	50.1892	31.2018	1.35	99.9	163	120
7	50.2320	31.6008	1.60	110.5	266	103
9	50.2471	31.0277	1.89	48.9	116	84
10	50.2496	30.8484	1.08	501.8	1460	368
11	50.2501	31.2212	2.03	20.4	83	74
12	50.3050	30.9977	1.98	110.6	915	70
13	50.3715	30.9517	1.09	175.6	243	175
14	50.4496	31.8914	1.78	109.9	229	79
15	50.4506	31.2420	1.01	250.4	226	249
16	50.5573	31.2059	1.52	122.0	132	108
17	50.5910	32.1084	0.93	−163.8	293	289
18	50.6189	29.6954	0.98	−499.8	447	349

Continued on next page



BDensitySensitivity.png

Figure 4. Sensitivity of the derived $B_{\parallel}$ distribution to the assumed initial gas density n_0 in the chemical evolution model. Box plots show the distribution of $|B_{\parallel}|$ for the ON-point sample at $n_0 = 500, 1000, 1500, 2000 \text{ cm}^{-3}$, with the fiducial model ($n_0 = 1000 \text{ cm}^{-3}$) highlighted. Higher n_0 yields higher electron column densities and thus lower inferred $|B_{\parallel}|$. The fractional change in $B_{\parallel}$ is approximately linear in n_0 over this range.



BTemperatureSensitivity.png

Figure 5. Sensitivity of the derived $B_{\parallel}$ distribution to the assumed gas kinetic temperature T_0 in the chemical evolution model. Box plots show the distribution of $|B_{\parallel}|$ at $T_0 = 8, 10, 12, 14, 16$ K, with the fiducial model ($T_0 = 12$ K) highlighted. The temperature dependence is weaker than the density dependence (Figure ??), with $B_{\parallel}$ values shifting by $\sim 10 - 20\%$ across the explored range.

Table 4 – Continued from previous page

ID	RA	Dec	A_V	$B_{\parallel}$	$\delta B_{\parallel}^+$	$\delta B_{\parallel}^-$
	(deg)	(deg)	(mag)	(μG)	(μG)	(μG)
19	50.6458	28.9755	2.09	-111.7	69	150
20	50.6463	29.1336	1.65	384.9	2454	216
21	50.6487	31.2004	2.17	17.0	169	57
22	50.6830	29.5650	1.94	-205.5	70	721
23	50.7428	31.0642	1.84	245.4	202	121
24	50.7831	29.1996	1.20	-299.9	237	313
25	50.8332	28.9076	1.63	-249.7	115	853
26	50.8355	32.1642	1.65	468.3	769	169
27	50.8542	30.0189	0.76	-903.1	857	427
28	50.8606	29.0470	1.70	219.8	161	113
29	50.8620	29.3801	2.31	-197.7	53	1377
30	50.8724	29.2855	1.80	-381.3	111	702
31	50.8965	28.9453	1.61	-86.0	104	106
32	50.9051	30.0765	2.20	-80.7	65	381
33	50.9078	29.5763	1.65	-252.8	153	158
34	50.9385	31.7498	1.53	-98.8	98	569
35	50.9428	29.3842	2.28	-177.2	65	812
36	50.9735	31.7548	1.58	-198.5	118	1179
37	50.9770	28.9037	1.95	143.6	1159	80
38	51.0397	30.1021	2.22	125.1	138	71
39	51.0460	30.1008	2.22	101.0	116	66
40	51.0462	31.0914	1.52	376.2	151	226
41	51.0504	29.3711	1.48	-305.4	174	227
42	51.0636	29.5809	1.14	-328.0	273	290
43	51.0755	30.7043	2.64	71.5	149	52
44	51.0822	30.7016	2.64	80.2	165	52
45	51.0981	30.8764	2.85	-2.0	52	52
46	51.1024	29.8281	1.90	294.1	245	120
47	51.1057	30.8731	2.85	13.0	51	49
48	51.1416	31.6834	1.71	-133.0	84	278
49	51.1639	31.7010	1.63	-115.2	96	722
50	51.1687	31.4791	1.74	54.5	102	86
51	51.1691	30.1096	2.05	-20.1	73	75
52	51.1747	28.7424	1.76	-45.1	85	99
53	51.1926	29.9887	3.25	-19.1	39	273
54	51.1937	30.8288	3.58	56.9	71	45
55	51.2013	30.8282	3.58	61.1	72	42
56	51.2293	28.7047	1.49	-1025.2	541	1418
57	51.2424	29.3694	1.44	377.6	143	211
58	51.2490	28.7534	1.44	43.7	124	123
59	51.2559	32.1766	1.27	-326.7	238	1315
60	51.2922	31.4768	2.14	-91.6	58	98
61	51.2987	28.7342	1.48	-367.0	205	114
62	51.3003	28.9870	2.08	31.2	122	80
63	51.3128	29.0920	1.67	239.5	442	116
64	51.3191	29.8315	1.76	62.0	118	101

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Table 4 – Continued from previous page

ID	RA	Dec	A_V	$B_{\parallel}$	$\delta B_{\parallel}^+$	$\delta B_{\parallel}^-$
	(deg)	(deg)	(mag)	(μG)	(μG)	(μG)
65	51.3228	31.2866	2.44	-87.5	64	76
66	51.3971	29.0263	1.09	-316.5	249	198
67	51.4262	30.1062	7.10	-47.8	33	38
68	51.4266	29.1665	0.80	-684.6	634	821
69	51.4340	31.9809	1.60	-266.8	140	520
70	51.4618	29.1460	0.93	-699.6	559	1364
71	51.4961	31.4853	2.76	-98.4	58	74
72	51.5023	28.7551	1.16	-256.6	232	241
73	51.5183	32.1704	1.72	293.0	1984	111
74	51.5235	28.9684	1.31	-190.1	167	828
75	51.5327	30.1748	4.28	-71.0	37	56
76	51.5744	31.4566	2.71	-48.2	51	49
77	51.5804	31.4549	2.71	-37.8	48	47
78	51.5841	29.0843	2.08	-48.4	72	168
79	51.5884	29.0827	2.08	-20.3	77	99
80	51.5943	29.3633	1.62	-65.0	95	161
81	51.6315	28.7747	1.41	-132.7	132	206
82	51.7813	29.8153	3.04	-137.8	57	134
83	51.7953	29.0226	1.75	-197.0	77	204
84	51.8040	29.7065	1.28	17.9	149	149
85	51.8170	29.9075	3.92	-62.8	36	94
86	51.8346	31.8038	2.09	-35.7	66	75
87	51.8352	31.8085	2.09	-21.6	64	67
88	51.8414	31.8042	2.09	-56.1	69	88
89	51.8452	31.1679	2.87	65.3	101	57
90	51.8739	30.2471	15.61	-22.1	29	30
91	51.9147	31.4026	2.87	-62.7	47	68
92	51.9258	30.1947	9.59	-30.3	29	31
93	51.9343	30.9495	1.97	93.9	99	81
94	51.9396	29.0660	2.54	-215.3	52	1167
95	52.0150	30.9776	2.73	28.0	59	59
97	52.0284	29.4441	1.16	-247.8	189	630
98	52.0941	29.0017	2.62	-194.4	52	181
99	52.0991	29.0052	2.62	-227.2	54	209
100	52.1048	30.7135	2.84	25.2	53	52
101	52.1113	29.0181	2.62	-223.5	60	208
102	52.1147	30.8918	3.81	87.2	50	47
103	52.1211	28.9815	2.10	-21.0	67	68
104	52.1263	30.8459	5.16	47.9	36	31
105	52.1315	30.8459	5.16	47.2	35	30
106	52.1388	30.8443	5.16	47.7	36	31
107	52.1503	29.3655	2.32	-78.3	50	130
108	52.1588	30.8360	4.21	50.6	35	32
109	52.1682	30.8317	4.21	48.8	34	32
110	52.1796	31.6535	4.46	184.6	216	46
111	52.2177	29.5093	2.50	-169.1	69	115

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Table 4 – Continued from previous page

ID	RA	Dec	A_V	$B_{\parallel}$	$\delta B_{\parallel}^+$	$\delta B_{\parallel}^-$
(deg)	(deg)	(deg)	(mag)	(μG)	(μG)	(μG)
112	52.2307	29.5287	2.90	-124.0	46	203
113	52.2943	31.5316	6.93	11.9	33	33
114	52.2989	31.0010	3.37	67.5	57	55
115	52.3346	31.1124	4.07	23.6	39	38
116	52.3786	31.3691	4.97	140.4	61	50
117	52.4963	30.6062	2.80	-39.7	56	62
118	52.4998	30.6105	2.64	-31.3	55	58
119	52.5153	30.5489	3.74	-91.2	50	100
120	52.5160	30.5547	3.74	-104.6	52	113
121	52.5203	31.1963	2.90	-122.4	68	168
122	52.5215	30.5514	3.74	-45.4	38	58
123	52.5238	30.6039	2.39	-46.3	64	93
124	52.5331	30.5480	3.75	-25.9	34	42
125	52.5379	30.5471	3.75	-24.5	34	41
126	52.5403	31.4094	2.05	-310.9	121	87
127	52.5764	30.9263	1.89	-374.6	223	158
128	52.5801	31.7726	2.77	-199.5	86	283
129	52.6025	31.7878	2.98	-142.3	60	223
130	52.6538	31.8737	2.56	-276.4	103	501
131	52.6616	31.5128	2.19	-183.9	90	93
132	52.6707	30.1919	0.79	-416.7	550	431
133	52.7403	31.5680	3.24	-32.9	57	60
134	52.7505	31.3681	1.93	-148.8	95	413
135	52.7948	31.6515	3.69	-147.7	50	93
136	52.7972	32.0584	1.49	-197.8	156	139
137	52.8051	31.9214	1.65	89.2	81	86
138	52.8248	30.7921	2.85	82.7	82	54
139	52.8296	30.7907	1.82	135.7	99	108
140	52.8428	31.0572	1.95	-109.9	90	319
141	52.8888	31.8530	1.64	153.8	118	117
142	52.9096	31.9073	1.99	21.9	76	74
143	52.9134	30.6760	2.12	-76.9	73	65
144	52.9144	30.6719	2.12	-66.9	66	65
145	52.9186	30.6755	2.12	-75.2	72	64
146	52.9761	31.0781	2.20	-301.5	148	127
147	53.0330	30.4347	1.78	58.6	112	111
148	53.0450	30.6622	2.26	55.2	79	75
149	53.0664	31.7577	1.78	118.6	86	91
150	53.0749	31.7560	1.69	108.0	100	89
151	53.1052	31.0868	3.46	44.3	50	38
152	53.1497	31.8378	1.73	352.6	581	117
153	53.1670	31.8621	1.50	300.4	449	122
154	53.1675	31.8530	1.50	310.5	464	124
155	53.1680	31.8338	1.67	179.5	272	110
156	53.1749	30.3356	2.27	-38.3	53	54
157	53.1795	31.8492	1.50	376.8	562	146

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Table 4 – Continued from previous page

ID	RA	Dec	A_V	$B_{\parallel}$	$\delta B_{\parallel}^+$	$\delta B_{\parallel}^-$
(deg)	(deg)	(deg)	(mag)	(μG)	(μG)	(μG)
158	53.1900	31.4131	3.90	34.6	52	50
159	53.2240	31.7203	2.17	62.0	89	58
160	53.2537	31.3384	5.32	-104.4	46	50
161	53.2750	30.8791	3.74	22.3	42	38
162	53.2860	30.8468	2.22	205.5	95	123
163	53.3239	30.7891	2.91	-3.4	64	64
164	53.3478	31.8360	1.06	566.2	1552	400
165	53.3771	30.1060	1.31	-31.0	122	164
166	53.3779	31.2996	10.37	-72.5	34	34
167	53.4622	30.3840	1.89	-68.5	98	467
168	53.4651	30.5202	1.69	107.0	148	98
169	53.4731	31.2204	4.28	-16.6	46	48
170	53.4811	29.8010	0.94	-504.4	408	476
171	53.4896	30.3951	1.53	-43.4	112	238
172	53.5077	31.3111	6.17	-35.8	29	30
173	53.5223	30.5026	1.14	-57.2	162	163
174	53.5459	30.3858	1.56	-246.7	127	1244
175	53.5491	31.0069	2.91	-63.0	46	74
176	53.5659	30.7473	1.31	489.9	1852	271
177	53.5710	31.2029	4.09	-55.1	32	43
178	53.5778	30.8547	1.58	-201.4	133	1199
179	53.5802	30.8507	1.58	-198.6	131	1182
180	53.6689	30.6153	1.19	272.9	255	257
181	53.7031	31.1146	2.54	36.1	53	54
182	53.7377	30.8237	2.13	102.6	76	72
183	53.7721	30.7852	2.04	-84.2	67	68
184	53.7787	30.7850	2.04	-109.1	71	74
187	53.8006	30.7847	1.70	61.8	78	82
188	53.8651	30.3989	0.85	195.7	378	299
189	53.8766	30.5400	2.11	-90.7	58	104
190	53.8765	30.7783	2.21	54.6	85	54
191	53.8769	30.6246	1.55	-134.4	90	91
192	53.8881	31.5752	1.65	168.3	109	109
193	53.8906	31.5713	1.65	145.8	103	103
194	53.9287	30.3202	1.74	-43.5	88	313
195	54.2008	29.9774	1.21	-201.9	161	756
196	54.3165	31.2493	3.92	184.4	58	43
200	54.4577	31.2542	2.42	358.6	52	110
201	54.4661	30.9337	3.08	83.2	77	38
202	54.5013	31.1029	2.45	158.4	113	59
203	54.6136	30.8844	2.38	153.7	99	76
204	54.9238	30.8482	1.71	301.2	225	145

5. DISCUSSION

[TODO: Discussion section to be written - will include interpretation of field reversal, comparison with plane-

of-sky field structure, implications for cloud formation scenarios, etc.]

6. CONCLUSIONS AND FUTURE WORK

[TODO: Conclusions to be written]

7. SOFTWARE

The MC-BLOS software is publicly available at: <https://github.com/MehrnooshTahani/MC-BLOS>. All data analysis and figure generation were performed using Python 3.9 with the following packages: NumPy (?), Pandas (?), Matplotlib (?), and Astropy (??).

8. ACKNOWLEDGMENT

[TODO: Add acknowledgments]